

Jorge Anais Vilchez

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Professional with a scientific background and extensive experience in technical astronomical environments acquiring, reducing, and analyzing astronomical data. Highly proficient in data processing, machine learning algorithms, and Python.

Selected Work & Research Experience

Astroinformatics Researcher <i>Universidad de Antofagasta & ESO</i>	2023-present
Engineer automated data pipelines to integrate and clean complex photometric datasets from international observatories, including wide-field data from the VVV/VVVX and Gaia DR3 surveys to map galactic structures.	
Assistant Professor <i>Duoc UC</i>	2023-present
Deliver lectures on Machine Learning, Deep Learning, Data Mining, and Big Data. Mentor students and spearhead instructional design for the Artificial Intelligence engineering track and online Data Science programs, integrating interdisciplinary applications such as data sonification.	
Data Scientist <i>Grupo Kaufmann AWTO</i>	2021-2022
Designed and implemented scalable machine learning models to optimize corporate mobility and vehicle-sharing operations.	
Research Assistant, Observer & Data Analyst <i>Various Institutions: NOIRLab, Las Campanas Observatory, UC Santa Cruz</i>	2014-2025
Served as a research assistant for international observing projects. Executed observations, data reduction, and image data analysis across multiple instruments, telescopes, and observatories.	

Education

Doctorate in Astrophysics and Astroinformatics <i>Universidad de Antofagasta</i>	2027 (expected)
Master in Astronomy <i>Universidad de Antofagasta</i>	2021
Licentiate in Astronomy <i>Pontificia Universidad Católica de Chile</i>	2014

Relevant Certifications & Training

Data Governance, Regulation, and Ethics. <i>Universidad de Chile</i>	2026
Deep Learning Specialization. <i>Coursera, DeepLearning.AI</i>	2025
Evolutionary Algorithms and Visual Recognition with Deep Learning. <i>Universidad de Chile</i>	2025
Transformers, Use of Agents, LangChain and RAG. <i>Universidad de Chile</i>	2025
Natural Language Processing. <i>Universidad de Chile</i>	2024

Skills

Languages: Spanish (native) and English.
Computer Languages & Frameworks: Python, PyTorch, Keras, Pandas, NumPy, SciPy, Scikit-learn, Plotly, Bash, R
Tools & Infrastructure: Docker, Git, PostgreSQL, Linux environments.
Specialties: Astroinformatics, Machine Learning, Exploratory Data Analysis, Technical Communication.
Publications: Co-authored 15 peer-reviewed publications focusing on statistical modeling, transient event detection, and extensive dataset calibration.